



Module 11 Day 2: Cryptography

Solution Guide: Cryptography Refresher

The goal of this exercise was to review how to use OpenSSL and illustrate the challenge that securely exchanged keys pose to security professionals.

- Write a message (you can use Nano or Vi) to create a message titled `meetingplace_update.txt` containing the following information:

Unset

Dear [partner's code name]

Captain Strickland would like us to now meet at Lou's Cafe at 6pm to discuss some additional information about the Alphabet Bandit. See you there!

From,

[your code name]

- Use OpenSSL to encrypt the message by running the following:
`openssl enc -pbkdf2 -nosalt -aes-256-cbc -in meetingplace_update.txt -out meetingplace_update.txt.enc -base64 -K <your_key> -iv <your_iv>`
- Send the following to your partner:
 - `meetingplace_update.txt.enc`
 - Key
 - IV
- Upon receiving your partner's message, key, and IV, decrypt the message with the following command, updated with your partner's key and IV in the correct

locations:

```
openssl enc -pbkdf2 -nosalt -aes-256-cbc -d -in
```

```
meetingplace_update.txt.enc -base64 -K [partner key] -iv [partner IV]
```

- It's likely that most of you used Slack or email to transmit the key and IV. Slack and email may seem secure; however:
 - Emails are typically not encrypted.
 - Slack administrators can have access to view private messages between parties.